

EUCLID ASCENT

ALGEBRA AND FUNCTIONS

MARK SCHEME

WRITTEN BY TMUA EUCLID

L01 / Q1

Rationalising the denominator gives

$$\frac{1}{\sqrt{5}-2} \cdot \frac{\sqrt{5}+2}{\sqrt{5}+2} = \frac{\sqrt{5}+2}{5-4} = \sqrt{5}+2.$$

ANSWER **A**

L01 / Q2

Since the roots of $x^2 + 4x - 1 = 0$ are p and q ,

$$p + q = -4, \quad pq = -1.$$

Therefore

$$(p+1)(q+1) = pq + p + q + 1 = -1 - 4 + 1 = -4.$$

ANSWER **C**

L01 / Q3

The product

$$(x-1)(x-4)$$

is negative strictly between its roots. Hence

$$1 < x < 4.$$

The integers in this interval are 2 and 3, so there are exactly two possible values.

ANSWER **F**

L02 / Q1

Set

$$u = |x-2|,$$

where $u \geq 0$. The equation becomes

$$u = u^2,$$

so

$$u = 0 \quad \text{or} \quad u = 1.$$

Thus $|x-2| = 0$ gives $x = 2$, while $|x-2| = 1$ gives $x = 1$ or $x = 3$. There are three real solutions.

ANSWER **F**

L02 / Q2

By the remainder theorem, the remainder when $(x + 1)P(x)$ is divided by $x - 2$ is its value at $x = 2$:

$$(2 + 1)P(2) = 3 \cdot 5 = 15.$$

ANSWER **E****L02 / Q3**

For $x \neq 1$,

$$\frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1} = x + 1.$$

At $x = 1$, however, the original fraction is undefined. The equality therefore holds for every real x except 1.

ANSWER **B****L03 / Q1**

Using

$$P(x + 1) - P(x) = 2x + 1$$

with $x = 0$,

$$P(1) - P(0) = 1.$$

Since $P(0) = 3$, this gives $P(1) = 4$. Now take $x = 1$:

$$P(2) - P(1) = 3,$$

and hence

$$P(2) = 4 + 3 = 7.$$

ANSWER **C****L03 / Q2**

Complete the square:

$$f(x) = (x - 2)^2 + 3.$$

The graph is symmetric about $x = 2$. Since 6 is four units to the right of the axis, the other input with the same function value is four units to its left:

$$a = 2 - 4 = -2.$$

ANSWER **D****L03 / Q3**

Square the given relation:

$$(2^x + 2^{-x})^2 = 9.$$

Expanding,

$$4^x + 2 + 4^{-x} = 9,$$

so

$$4^x + 4^{-x} = 7.$$

ANSWER **B**

L04 / Q1

Since $P(1) = P(3)$, the axis of symmetry of the quadratic is $x = 2$. Write

$$P(x) = a(x - 2)^2 + c.$$

Using $P(1) = 5$ and $P(0) = 8$,

$$a + c = 5, \quad 4a + c = 8.$$

Subtracting gives $3a = 3$, so $a = 1$ and then $c = 4$. Therefore

$$P(2) = c = 4.$$

ANSWER **E**

L04 / Q2

The remainder after division by $x^2 - 1$ has degree at most one, so write it as

$$ax + b.$$

At $x = 1$ and $x = -1$, the divisor is zero, so the given values yield

$$a + b = 2, \quad -a + b = -2.$$

Adding gives $b = 0$, and hence $a = 2$. The remainder is therefore

$$2x.$$

ANSWER **A**

L04 / Q3

Let $u = x + 2$. Then

$$\frac{2x + 4}{x^2 + 4x + 8} = \frac{2u}{u^2 + 4}.$$

Since

$$(u - 2)^2 \geq 0,$$

we have $u^2 + 4 \geq 4u$. As the denominator is positive,

$$\frac{2u}{u^2 + 4} \leq \frac{1}{2}.$$

Equality occurs when $u = 2$, corresponding to $x = 0$. Thus the greatest possible value is

$$\frac{1}{2}.$$

ANSWER **D****L05 / Q1**

For every real x ,

$$P(x)^2 = P(x),$$

so

$$P(x)(P(x) - 1) = 0.$$

Hence $P(x)$ can take only the values 0 and 1. Since a polynomial is continuous, it cannot change between these two values without taking every intermediate value. Therefore it must be constant.

Consequently,

$$P(x) \equiv 0 \quad \text{or} \quad P(x) \equiv 1.$$

ANSWER **D****L05 / Q2**

Starting from

$$x = 2 + \frac{x}{2x+1},$$

multiply by $2x+1$:

$$(x-2)(2x+1) = x.$$

Expanding and simplifying gives

$$x^2 - 2x - 1 = 0.$$

Therefore

$$x = 1 \pm \sqrt{2}.$$

Only the positive value is permitted, so

$$x = 1 + \sqrt{2}.$$

ANSWER **C****L05 / Q3**

Let

$$p = xy.$$

Since $x + y = xy$, we also have $x + y = p$. Therefore

$$x^2 + y^2 = (x + y)^2 - 2xy = p^2 - 2p.$$

Using $x^2 + y^2 = 10$,

$$p^2 - 2p - 10 = 0,$$

so

$$p = 1 \pm \sqrt{11}.$$

Because x and y are positive, $p = xy > 0$. Hence

$$xy = 1 + \sqrt{11}.$$

ANSWER **B**

L06 / Q1

For positive x , the equation becomes

$$|x^2 - 4| = x.$$

If $x^2 \geq 4$, then

$$x^2 - 4 = x,$$

whose positive solution is

$$p = \frac{1 + \sqrt{17}}{2}.$$

If $x^2 < 4$, then

$$4 - x^2 = x,$$

whose positive solution is

$$q = \frac{-1 + \sqrt{17}}{2}.$$

Both solutions satisfy their respective case conditions. Therefore

$$p + q = \sqrt{17}.$$

ANSWER **C**

L06 / Q2

Complete the square in the numerator and denominator:

$$f(x) = \frac{(x-2)^2 - 1}{(x-2)^2 + 1}.$$

Let

$$t = (x-2)^2,$$

so $t \geq 0$. Then

$$f(x) = \frac{t-1}{t+1} = 1 - \frac{2}{t+1}.$$

The least value occurs at $t = 0$, giving -1 . As t increases, the expression approaches 1 but never reaches it. Thus the range is

$$[-1, 1).$$

ANSWER **E**

L06 / Q3

Write

$$P(x) = ax^2 + bx + c.$$

The leading coefficient of $P(P(x))$ is a^3 . Comparing it with the leading coefficient of $x^4 + 2x^2 + 2$ gives

$$a^3 = 1,$$

so $a = 1$. The coefficient of x^3 in the composition is then $2b$, and therefore $b = 0$.

Thus

$$P(x) = x^2 + c,$$

and

$$P(P(x)) = (x^2 + c)^2 + c.$$

Comparing the coefficient of x^2 gives $2c = 2$, so $c = 1$. Hence

$$P(1) = 2.$$

ANSWER **A**

L07 / Q1

Let

$$y = f(x).$$

Then $f(f(x)) = f(x)$ becomes

$$f(y) = y.$$

Since $f(y) = (y - 2)^2 + 1$,

$$(y - 2)^2 + 1 = y,$$

so

$$y^2 - 5y + 5 = 0.$$

Its two roots are

$$y = \frac{5 \pm \sqrt{5}}{2},$$

and both are greater than 1. For either value of y , the equation

$$(x - 2)^2 + 1 = y$$

has two distinct real solutions. The total number of real solutions is therefore

$$2 + 2 = 4.$$

ANSWER **A**

L07 / Q2

Factorise:

$$x^4 - kx^3 + kx - 1 = (x^2 - 1)(x^2 - kx + 1).$$

For $x > 1$, the first factor is positive. Hence the original expression is positive for every $x > 1$ exactly when

$$x^2 - kx + 1 > 0$$

for every $x > 1$. Dividing by $x > 0$, this is

$$x + \frac{1}{x} > k.$$

For $x > 1$, the left-hand side is always greater than 2, but can be arbitrarily close to 2. Therefore the required condition is

$$k \leq 2.$$

ANSWER **B**

L07 / Q3

From $f(3) = 7$,

$$\frac{3a + b}{3c - a} = 7.$$

Therefore

$$3a + b = 21c - 7a,$$

so

$$b = 21c - 10a.$$

Now

$$f(7) = \frac{7a + b}{7c - a} = \frac{21c - 3a}{7c - a} = \frac{3(7c - a)}{7c - a} = 3.$$

The question states that $f(7)$ is defined, so $7c - a \neq 0$.

ANSWER **D**

L08 / Q1

Each quadratic $x^2 + px + 1$ and $x^2 + qx + 1$ has two distinct positive roots.

For $x^2 + px + 1$, the sum of the roots is $-p$, so positivity requires $p < 0$. Two distinct real roots also require

$$p^2 > 4.$$

Together these conditions give $p < -2$. Similarly,

$$q < -2.$$

On expanding the product of the two quadratics, the coefficient in question is

$$b = pq + 2.$$

Since $p < -2$ and $q < -2$, we have $pq > 4$, and hence

$$b > 6.$$

ANSWER **F**

L08 / Q2

Suppose the expression takes the value y . Then

$$x^2 + 6x + 1 = y(x^2 + 2x + 5),$$

or

$$(1 - y)x^2 + (6 - 2y)x + (1 - 5y) = 0.$$

For a real value of x to exist, the discriminant must be non-negative:

$$(6 - 2y)^2 - 4(1 - y)(1 - 5y) \geq 0.$$

Simplifying,

$$16(2 - y^2) \geq 0,$$

so

$$-\sqrt{2} \leq y \leq \sqrt{2}.$$

The endpoint values occur when the quadratic in x has a repeated real root. Hence the greatest possible value is

$$\sqrt{2}.$$

ANSWER E

L08 / Q3

Expanding the product gives

$$(x - u)(x - v)(x - w) = x^3 - (u + v + w)x^2 + (uv + uw + vw)x - uvw.$$

Comparing coefficients with

$$x^3 - ux^2 + vx - w$$

gives

$$u + v + w = u, \quad uv + uw + vw = v, \quad uvw = w.$$

Since $w \neq 0$, the last equation gives $uv = 1$. The first gives $v + w = 0$, so $w = -v$.

Substituting these into the middle equation,

$$-v^2 = v.$$

As $v \neq 0$, it follows that $v = -1$. Hence $w = 1$, and $uv = 1$ gives $u = -1$. Therefore

$$u + v + w = -1.$$

ANSWER B

L09 / Q1

Set

$$y = x^2 - 4x = (x - 2)^2 - 4.$$

Thus $y \geq -4$, and the equation becomes

$$y^2 - ky + 4k = 0.$$

Each root $y > -4$ produces two distinct real values of x , while the root $y = -4$ produces the single value $x = 2$. Therefore exactly three distinct real solutions require one y -root to equal -4 and the other to be greater than -4 .

Substituting $y = -4$,

$$16 + 4k + 4k = 0,$$

so

$$k = -2.$$

The other y -root is then 2, which produces two further values of x . Hence the required value is

$$k = -2.$$

ANSWER C

L09 / Q2

Write

$$P(x) = x^2 + ux + v,$$

so $P(0) = v$. The two roots are given as

$$P(1) = 1 + u + v, \quad P(-1) = 1 - u + v.$$

Their sum must equal $-u$, so

$$2 + 2v = -u.$$

Their product must equal v , so

$$(1 + v)^2 - u^2 = v.$$

Substituting $u = -2 - 2v$ gives

$$3v^2 + 7v + 3 = 0.$$

The product of the two possible values of $v = P(0)$ is therefore

$$\frac{3}{3} = 1.$$

ANSWER **E**

L09 / Q3

Define

$$Q(x) = (x + 1)P(x) - 1.$$

Since P has degree at most five, Q has degree at most six. The given values imply

$$Q(0) = Q(1) = \dots = Q(5) = 0.$$

Therefore

$$Q(x) = C x(x - 1)(x - 2)(x - 3)(x - 4)(x - 5)$$

for some constant C .

At $x = -1$,

$$Q(-1) = -1 = 720C,$$

so

$$C = -\frac{1}{720}.$$

At $x = 6$,

$$Q(6) = C \cdot 720 = -1.$$

But $Q(6) = 7P(6) - 1$, so

$$7P(6) - 1 = -1,$$

and hence

$$P(6) = 0.$$

ANSWER **F**